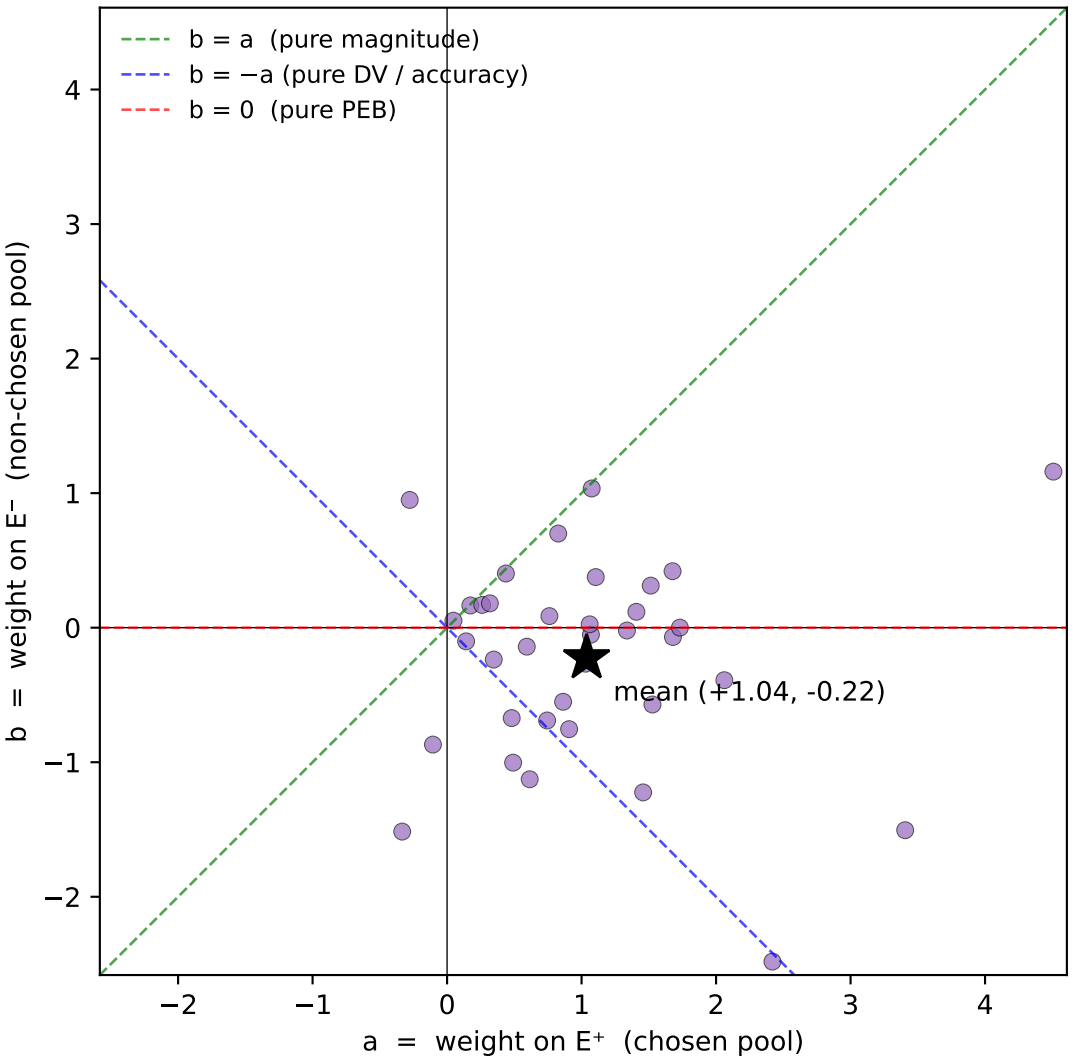
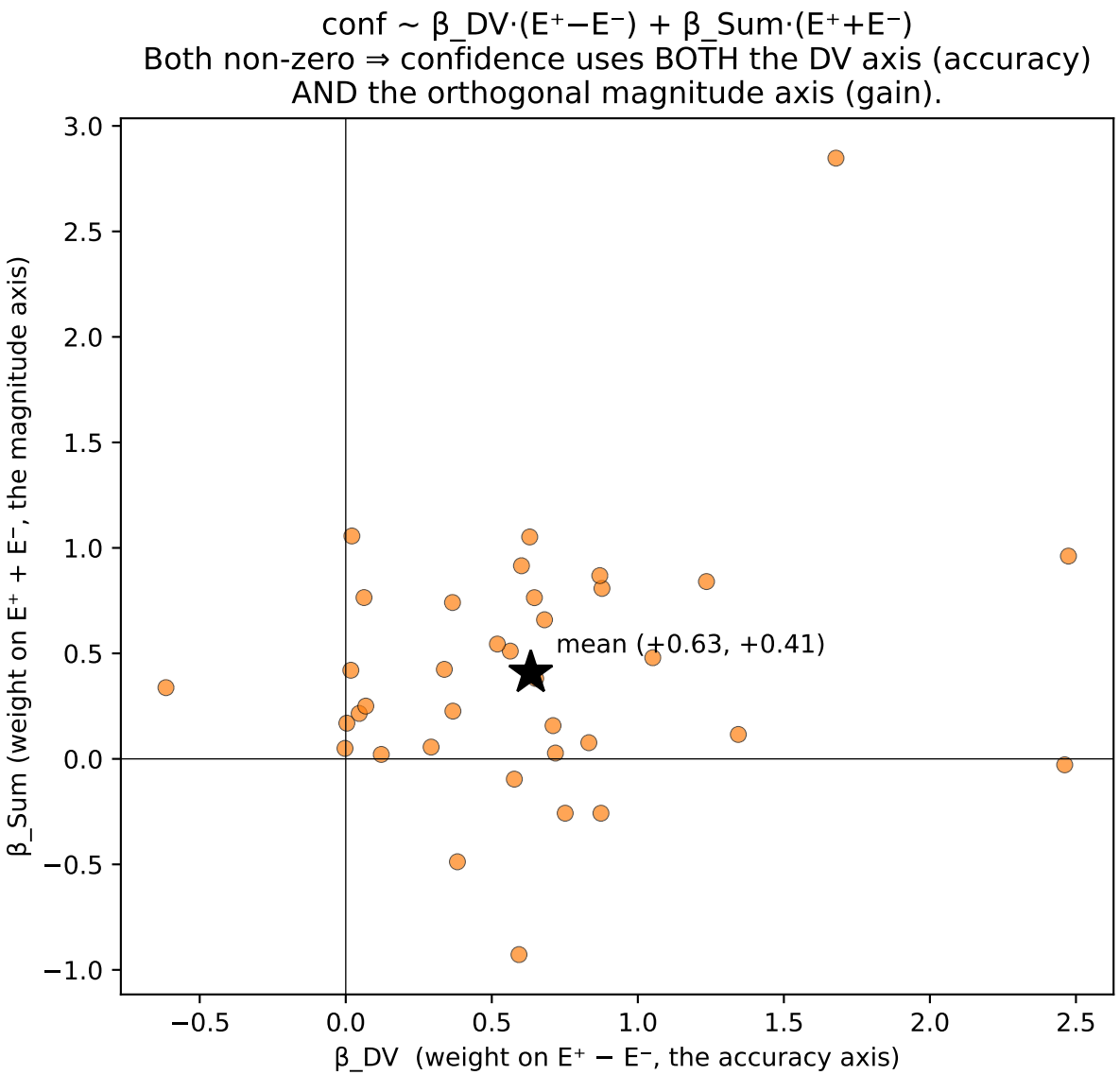
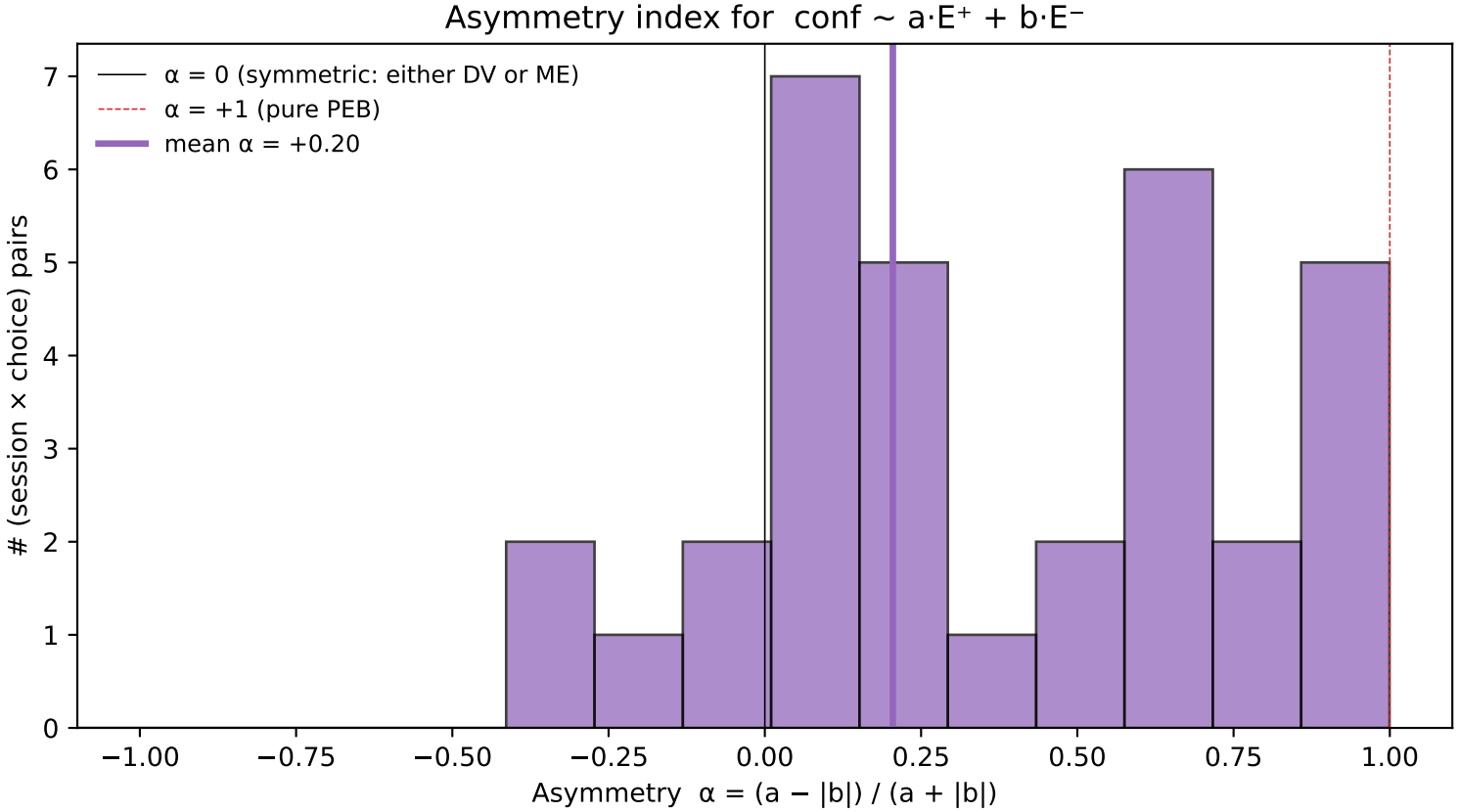

conf ~ a·E⁺ + b·E⁻ (one dot per session × choice)
Where in (a, b) space do the data sit?
b = a (pure magnitude)
b = −a (pure DV / accuracy)
b = 0 (pure PEB)
mean (+1.04, -0.22)
a = weight on E⁺ (chosen pool)
b = weight on E⁻ (non-chosen pool)
−2
−1
0
1
2
3
4


conf ~ β_DV·(E⁺−E⁻) + β_Sum·(E⁺+E⁻)
Both non-zero ⇒ confidence uses BOTH the DV axis (accuracy)
AND the orthogonal magnitude axis (gain).
mean (+0.63, +0.41)
β_Sum (weight on E⁺ + E⁻, the magnitude axis)
β_DV (weight on E⁺ − E⁻, the accuracy axis)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5


Asymmetry index for conf ~ a·E⁺ + b·E⁻
α = 0 (symmetric: either DV or ME)
α = +1 (pure PEB)
mean α = +0.20
# (session × choice) pairs
0
1
2
3
4
5
6
7
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Asymmetry α = (a − |b|) / (a + |b|)

```
PEB vs ME — population decomposition (29 sessions, 58 pairs after pref labels)

conf ~ a · E⁺ + b · E⁻   (logistic, E± z-scored within sess×choice)
  a (chosen pool)      mean = +1.038   p vs 0 = 4.92e-09
  b (non-chosen pool)  mean = -0.225   p vs 0 = 1.87e-01
  Wilcoxon (a+b ≠ 0):  mean = +0.813   p = 6.09e-05

Asymmetry α = (a − |b|)/(a + |b|):  mean = +0.205
  α ≈ +1 → pure PEB,   α ≈ 0 → symmetric (DV if b<0, ME if b>0)

Same model in DV / Sum basis:
  β_DV   (accuracy axis)  = +0.633   p = 1.08e-08
  β_Sum  (magnitude axis) = +0.408   p = 6.09e-05

Interpretation rules:
  pure DV / accuracy           a > 0,  b = −a   (sum a+b = 0)
  positive-evidence bias       a > 0,  b ≈ 0    (α near +1)
  magnitude / gain effect      a = b > 0        (α near 0,  with b > 0)

Reading off the data:
  Sign of a:   + (positive)
  Sign of b:   − (negative)
  Sign of (a+b):  +   →  excess magnitude component  (i.e. confidence ≠ pure DV)
```